The TNT and TNT3c recordings: what is measured

> **QUESTION**
>
> What do the TNT and TNT3c recordings contain, and which numbers are taken from them for the comparison with the model?

INPUTS

- `TNT_ovlp16_resp_perfly.pkl` and `TNT3c_ovlp16_resp_perfly.pkl` (Yang, 2026-10-15): per recording, a table of units × 16 odors
- `TNT_ovlp16_cosine_mean_RDM.csv`, `TNT3c_ovlp16_cosine_mean_RDM.csv`: Yang's mean RDMs, used as a check

OUTPUTS

- `odor_weighted/TNT_vs_model_per_odor.csv`, `TNT_vs_model_odor_pairs.csv` (data columns)
- `TNT_data_checks.csv` / `.json`, `TNT_rank_order.csv` (data rows)

SCRIPTS

- `scripts/tnt_compare.py` (section "data" and the helpers)
- `scripts/tnt_data_checks.py`
- `scripts/tnt_rank_order.py`
- `scripts/reports/r06_tnt_data.py` (this report)

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Yang recorded KC responses to 16 odors in two genotypes. In **TNT**, APL's synaptic output is blocked (tetanus toxin light chain expressed in APL). **TNT3c** is the control, with APL intact (Yang, 2026-10-15). This report describes the files, the quantities taken from them, and how reliable those quantities are. The pickles are read with a restricted loader (`scripts/orn_load.py`) that only accepts pandas and numpy classes.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
c	Condition: TNT (APL blocked) or TNT3c (control, APL intact)	
n	A recording. Each recording is one hemisphere and is treated as independent (Yang, 2026-10-15)	8 TNT, 11 TNT3c
i	A unit (one row of a recording's table): a segmented KC response region	1,638–2,462 per recording
o, a, b	Odors	16, the odors of the ORN table
$r_{io}^{(n)}$	Response of unit i in recording n to odor o, as provided	ΔF/F-like (units not stated); −2.6 to 7.4
$\phi_o^{(n)}$	Fraction of recording n's units with $r_{io}^{(n)} > 1$ ("responding")	0 to 1
ϕ_o^c	Mean of $\phi_o^{(n)}$ over the recordings of condition c	0 to 1
$S_{ab}^{(n)}$	Cosine similarity of odors a and b over the units of recording n	−1 to 1
S_{ab}^c	Mean of $S_{ab}^{(n)}$ over the recordings of condition c (= 1 − Yang's RDM)	−1 to 1

Symbol	What it stands for	Values, units, range
ΔS_{ab}	Effect of blocking APL on the pair: $S_{ab}^{\mathrm{TNT}} - S_{ab}^{\mathrm{TNT3c}}$	positive = more similar when APL is blocked
s_o	Suppression of odor o by APL: $\phi_o^{\mathrm{TNT}} / \phi_o^{\mathrm{TNT3c}}$	≥ 1 means more units respond when APL is blocked
$\mu,\ e_a$	Additive fit of ΔS: mean change μ and per-odor effect e_a (written f_a in the notes)	Section 5
ρ_s	Spearman rank correlation	−1 to 1

1 The recordings

There are 8 TNT recordings from 7 flies and 11 TNT3c recordings from 8 flies. A recording name reads `date_fly` + `FL`/`FR` (left or right side) + `fun` and a number. Where one fly gives two recordings, they are its two hemispheres. The label 20240802_5 appears in both files but names two different flies (Yang). Every recording is a table with one row per unit and one column per odor; the 16 odors are the same, in the same order, as in the ORN table.

Table 2. The recordings. "values > 1" is the share of all unit × odor responses above 1, the "responding" criterion used throughout.

condition	recording	side	units	values > 1
TNT3c (control, APL intact)	`20240710_2FRfun1`	R	1,952	9%
TNT3c (control, APL intact)	`20240712_2FLfun1`	L	1,947	8%
TNT3c (control, APL intact)	`20240726_4FRfun1`	R	1,638	8%
TNT3c (control, APL intact)	`20240801_3FLfun1`	L	1,956	7%
TNT3c (control, APL intact)	`20240801_3FRfun1`	R	2,070	5%
TNT3c (control, APL intact)	`20240809_3FRfun1`	R	1,961	6%
TNT3c (control, APL intact)	`20240710_2FLfun1`	L	1,790	7%
TNT3c (control, APL intact)	`20240809_3FLfun2`	L	2,219	4%
TNT3c (control, APL intact)	`20240814_1FLfun1`	L	2,350	10%
TNT3c (control, APL intact)	`20240802_5FRfun2`	R	2,374	5%
TNT3c (control, APL intact)	`20240830_3FRfun2`	R	1,894	6%
TNT (APL blocked)	`20240717_2FLfun1`	L	2,233	16%
TNT (APL blocked)	`20240717_3FLfun1`	L	2,462	19%
TNT (APL blocked)	`20240725_1FLfun1`	L	1,883	16%
TNT (APL blocked)	`20240801_4FRfun1`	R	1,728	15%
TNT (APL blocked)	`20240802_5FLfun1`	L	1,812	13%
TNT (APL blocked)	`20240807_3FLfun1`	L	2,253	14%
TNT (APL blocked)	`20240814_3FLfun1`	L	1,880	14%
TNT (APL blocked)	`20240814_3FRfun2`	R	2,238	15%

2 Response values and the "responding" criterion

The file does not state the units of $r_{io}^{(n)}$. The values behave like ΔF/F: most lie near 0, and a minority are large. A unit **responds** to an odor if its value is above 1. This is the threshold used in all comparisons (0.5 is checked as an alternative):

$$\phi_o^{(n)} = \frac{1}{N_n} \#\{\, i : r_{io}^{(n)} > 1 \,\}, \qquad \phi_o^{c} = \frac{1}{|c|} \sum_{n \in c} \phi_o^{(n)} \tag{1}$$

N_n = units in recording n; $|c|$ = recordings of condition c; every recording counts equally

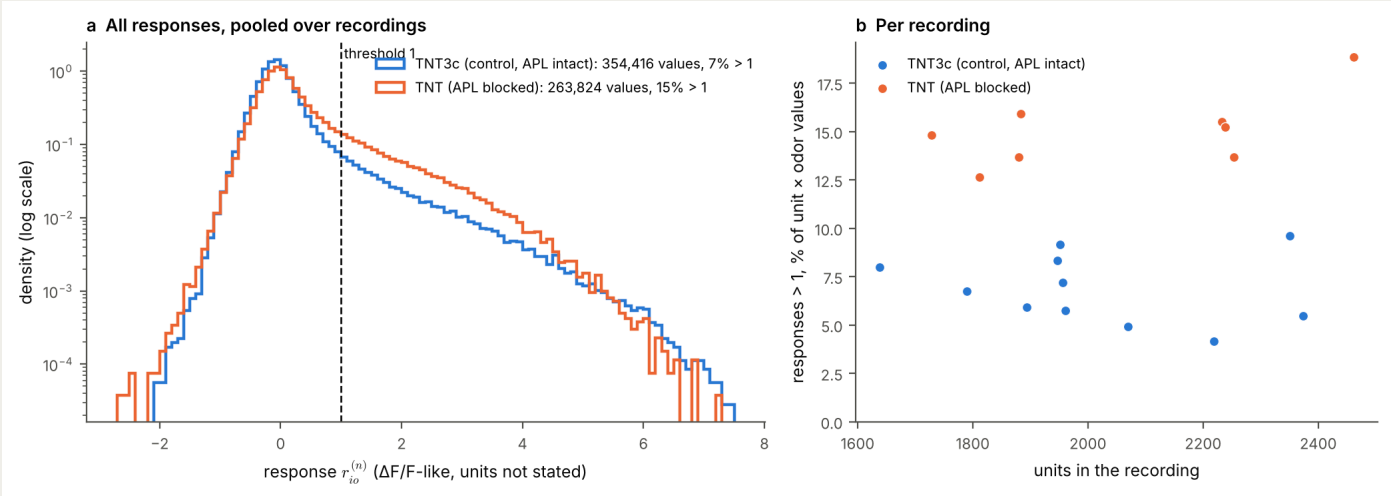


Figure 1. Response values in the two conditions. a, Distribution of every response value (unit × odor × recording) in each condition, as a density on a log scale. The dashed line is the responding threshold of 1. TNT has a heavier right tail. **b**, Each point is one recording: its number of units (horizontal) and the percentage of its unit × odor values above 1 (vertical).

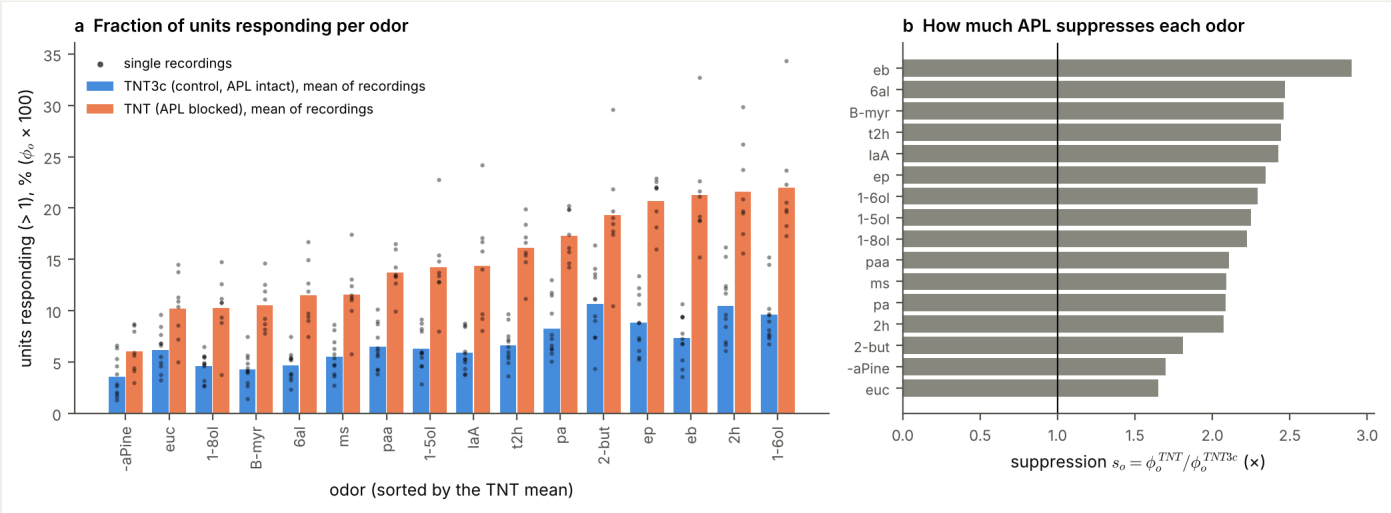


Figure 2. Fraction responding per odor, and suppression. a, Percentage of units responding (> 1) to each odor. Bars are the mean over recordings (blue TNT3c, orange TNT); black dots are single recordings. **b**, Suppression per odor: the TNT mean divided by the TNT3c mean. A value of 2 means twice as many units respond when APL is blocked.

$$s_o = \frac{\phi_o^{\mathrm{TNT}}}{\phi_o^{\mathrm{TNT3c}}} \tag{2}$$

On average over odors, 15% of units respond in TNT and 7% in TNT3c. Suppression ranges from ×1.65 (euc) to ×2.90 (eb). Report 07 uses these per-odor fractions to pin the model: with APL blocked, the model responds to each odor in the TNT fraction, and with APL intact its mean equals the TNT3c mean.

3 Odor-pair similarity

For each recording, the similarity of two odors is the cosine of their response vectors over all of its units (raw values, not thresholded). The condition's similarity is the mean over its recordings:

$$S_{ab}^{(n)} = \frac{\sum_i r_{ia}^{(n)}\, r_{ib}^{(n)}}{\sqrt{\sum_i \left(r_{ia}^{(n)}\right)^2}\,\sqrt{\sum_i \left(r_{ib}^{(n)}\right)^2}}, \qquad S_{ab}^{c} = \frac{1}{|c|}\sum_{n \in c} S_{ab}^{(n)} \tag{3}$$

$1 - S_{ab}^{c}$ is exactly Yang's RDM for that condition (checked to 10^{-6} for every pair). The effect of blocking APL on a pair is

$$\Delta S_{ab} = S_{ab}^{\mathrm{TNT}} - S_{ab}^{\mathrm{TNT3c}} \tag{4}$$

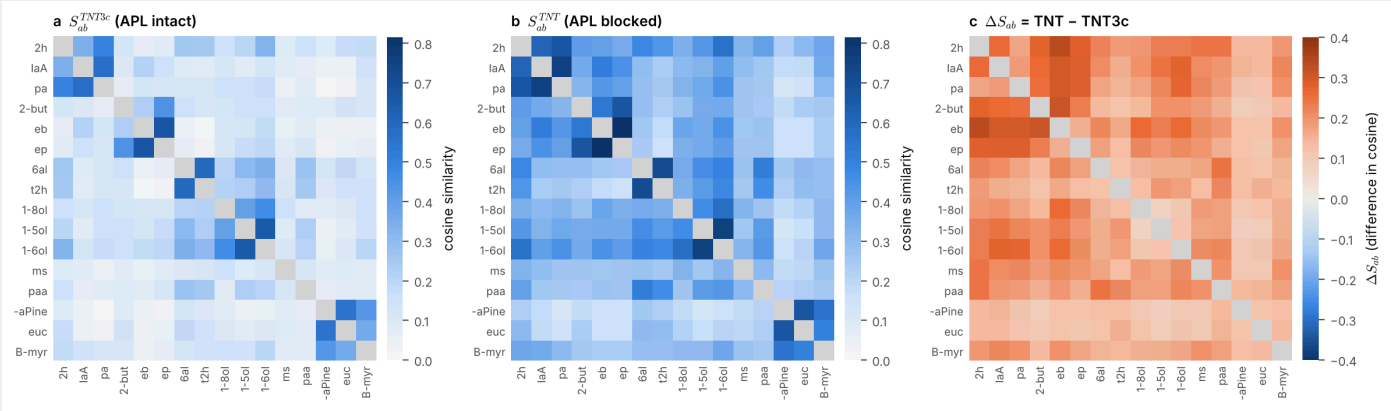


Figure 3. Odor-pair similarity in the two conditions. a, **b**, Mean cosine similarity of the units' responses for every odor pair, in TNT3c and TNT, on one scale (white 0, dark blue the highest value; slightly negative values, such as pa–-aPine in TNT3c, also show as white). **c**, The change, on a diverging scale (orange = more similar in TNT). Every pair is orange: blocking APL makes all 120 pairs more similar. Diagonal cells are left blank.

All 120 pairs are more similar in TNT. Mean similarity rises from 0.16 to 0.34 (mean ΔS = +0.178, range +0.07 to +0.33).

4 Splitting the change into per-odor and pair-specific parts

Some odors gain similarity with every partner. To separate that from what is specific to a pair, the change is fitted by least squares as a mean plus one effect per odor:

$$\Delta S_{ab} \approx \mu + e_a + e_b, \qquad \sum_a e_a = 0 \tag{5}$$

μ = mean change; e_a = how much more the pairs of odor a change than the average pair

R^2 of this fit is 0.65: 65% of the variation of ΔS across pairs is per odor. The residuals, $\Delta S_{ab} - (\mu + e_a + e_b)$, are the pair-specific part.

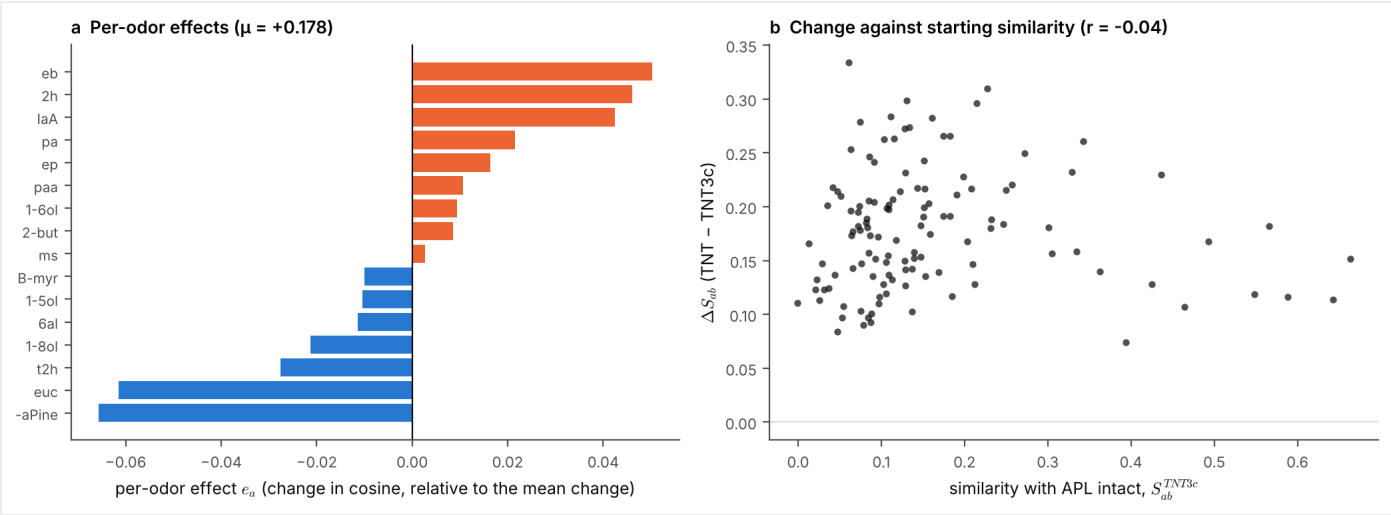


Figure 4. Which odors drive the change. a, The per-odor effect e_a of the additive fit, sorted. Orange: the odor's pairs rise more than average when APL is blocked. Blue: less. **b**, Each point is one odor pair: its similarity in TNT3c (horizontal) against its change (vertical).

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5 How reliable these numbers are

The recordings are the unit of resampling everywhere: bootstrap draws resample recordings with replacement within each condition, and split-half tests divide each condition's recordings into two random halves. A split-half correlation describes half the data. It is turned into an estimate for all the data with the Spearman–Brown formula, $r_{\text{full}} = 2r_{\text{half}}/(1 + r_{\text{half}})$.

- **Rank order of similarity.** Over the 120 pairs, S^{TNT} and S^{TNT3c} have ρ_s = 0.87. With half the recordings, two halves of the same condition give 0.90 (TNT3c) and 0.91 (TNT), while TNT vs TNT3c gives 0.83. The between-condition value is lower in 97.5% of splits. Dividing the between-condition value by the within-condition one gives about 0.92: the rank correlation expected without measurement noise. Report 09 and the local model use this as the data's value.
- **Change per pair.** The split-half reliability of ΔS caps any model's correlation with it near r = 0.84 (Report 07).
- **Duplicates.** No unit is repeated within or across recordings. Units whose nearest neighbour in their own recording is closer than expected are rare (0.5% of responsive TNT units, 1.6% TNT3c).
- **Two hemispheres of one fly.** Weighting each fly once instead of each recording changes the mean ΔS from 0.178 to 0.182, and the pattern over pairs correlates 0.99.

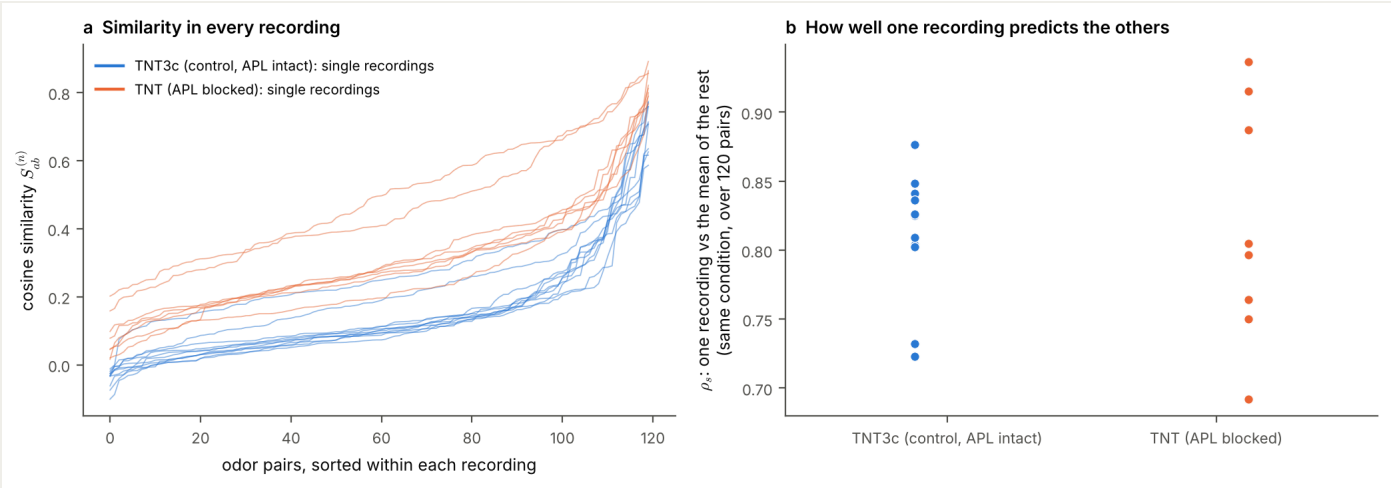


Figure 5. Consistency across recordings. a, Each line is one recording: its 120 pair similarities sorted from lowest to highest. TNT recordings (orange) sit higher throughout. **b**, Each point is one recording: the Spearman correlation of its pair similarities with the mean of the other recordings of the same condition. It measures how consistent the rank order of pairs is from hemisphere to hemisphere.

Caveats

The units are not matched across recordings or conditions, so every comparison is between condition means.

Response units are not stated. Thresholding at 1 assumes the values are on a comparable scale in both genotypes.

Noise lowers cosine similarity more when responses are small. TNT3c has smaller responses, so part of the rise in TNT may be noise. Thresholding at 1 before the cosine gives the same picture (every pair still rises; pattern r 0.94 with the raw change).

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